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Universe and Solar System

Big Bang Theory

- This theory is related to the origin of the universe.
- This theory was propounded by Georges Lemaitre in the year 1927.
- According to this theory, in the beginning all the matter from which the universe was created was located at one place in the form of a very small sphere (single atom).
- Due to excessive concentration, the point suddenly exploded. This caused the scattering of matter, which led to the origin of the universe
- This event of Big Bang happened 13.7 billion years ago.
- The solar system developed approximately 4.5 billion years ago.

Galaxy

- A group of stars which appears fuzzy. This is a mass of gas at the beginning of the star formation process.
- Our earth has its own galaxy which is called Milky Way. It is in spiral form.

The Origin & Destruction of Stars

- The rotation of the galaxy affects the clouds of gases present in the universe. Due to mutual gravity, nuclear fusion begins in their center. The process of hydrogen turning into helium begins, in this state it becomes a star.
- Due to the depletion of hydrogen in the centre, the central part of the star becomes compressed and hot. Slowly the star cools down and starts appearing red in colour.
- If the mass of a star is less than 1.4 MS (the mass of the Sun) then it loses its nuclear energy and turns into a white dwarf.
- The white dwarf cools down and transforms into a black dwarf.
- 1.4 Ms is called the Chandrashekhar limit. When the mass is more than this, the freely moving electrons leave the nucleus due to high velocity.
- Neutrons remain, this state is called pulsar. This also keeps on shrinking for an unlimited time and gets concentrated at a point, such a body containing matter of unlimited density is called a black hole.

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Constellation

A group of stars visible in the sky, which appear to us to be arranged in a certain shape.

Solar System

The group of various planets, asteroids, comets, meteors and other celestial bodies revolving around the Sun is called the solar system.

Planets

- Those celestial bodies which revolve around the Sun and have sufficient gravitational force are called planets.
- At present the number of planets in the solar system is 8.

Order of planets according to distance from the sun

Mercury -- Venus -- Earth -- Mars -- Jupiter -- Saturn -- Uranus -- Neptune

Order of planets according to mass

Mercury -- Mars -- Venus -- Earth -- Uranus -- Neptune -- Saturn -- Jupiter

Order of planets according to their period of revolution

Mercury -- Venus -- Earth -- Mars -- Jupiter -- Saturn -- Uranus -- Neptune

- Mercury and Venus do not have any satellites.They are also called Inner Planets.
- Venus is known as the 'morning star'. It is the second closest planet to the Sun. It is the brightest planet. It is called the twin sister of the Earth.
- The brightest planet in the solar system - Venus
- Slowest planet in the solar system - Neptune
- Mars is called the red planet. Mars has two poles like the Earth and it also rotates on its axis once in 24 hours. There are indications of suitability for life on this planet like on Earth.
- Rings are found around Jupiter and Saturn. The rings found around Jupiter are made of silicate.
- Saturn is the second largest planet in the solar system. Saturn's largest satellite is 'Titan'. Titan is equal in size to Mercury. It is the only satellite that has its own dense atmosphere like Earth.
- Asteroids revolve around the Sun between the orbits of Mars and Jupiter. They are small in size.
- The largest planet of the solar system - Jupiter
- The smallest planet in the solar system - Mercury

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Sun

- The Sun is a gaseous sphere consisting of 71% hydrogen, 26.5% helium and 2.5% other elements.
- The central part of the Sun is called the core, whose temperature is 1.5 x 10^7°C. The substances found in the center are in the form of gas and plasma due to high temperature.
- The temperature of the outer surface of the Sun is 6000°C. At the centre of the Sun, four hydrogen nuclei combine to form a helium nucleus. Nuclear fusion takes place at the centre of the Sun, which is the source of the Sun's energy.
- In case of a solar eclipse, the moon comes between the sun and the earth due to which the sunlight does not reach the earth. The part of the sun visible during a solar eclipse is called the 'sun corona'. This is the outermost layer of the sun

Perihelion & Aphelion

- **Perihelion** - When the earth is very close to the sun. This situation occurs on 3 January. In this situation, the distance between the earth and the sun is 15.21 crore km.
- **Aphelion** - When the earth is at the maximum distance from the sun. This situation occurs on 4th July. In this situation, the distance between the earth and the sun is 16 crore km.

Earth

- It is the fifth largest planet in size. The Earth's axis is inclined 23½^0 to the perpendicular to its orbital plane.
- This is the only planet in the solar system on which there is life. It has been called the Green Planet.
- The only satellite of the Earth is the Moon. Like the liquid interior of the Earth, the interior of the Moon is like a slime.
- The equatorial diameter of the Earth is 12756 km and the polar diameter is 12713.6 km. The average diameter of the Earth is 12742 km. It was first calculated by Eratosthenes.
- It takes about 8 minutes 20 seconds for light from the sun to reach the earth.
- The interior of the earth is divided into three parts- (1).Earth's crust (2).Mantle (3).Core
- The most abundant elements found on the Earth's crust are oxygen (46.60%), silicon (27.7%), aluminium (8.13%) and iron (5%).
- The average density of the Earth is 5.5gm/cm^3 and the average radius is 6370 km.
- As we go down into the earth, the temperature increases by 1^{0}c per 32m depth.

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- The distance of the moon from the earth is 384400 km.
- When an object is taken from the earth to the moon, its weight decreases because the gravitational acceleration of the moon is ⅙ as compared to the earth.
- There is no atmosphere on the moon

Latitude and Longitude Lines

Latitude

- The angular distance of a place situated north or south of the equator is called latitude.
- The equator represents 0^0 latitude. When drawn at an interval of 1^0 in the north and south direction of the equator, there are 90-90 latitude circles.
- A set of imaginary lines drawn around the Earth from east to west parallel to the equator are called latitude lines.
- The distance between two latitude lines is 111 km.

Longitude

- The angular distance of a place from the Greenwich line is called the longitude of that place.
- The imaginary line joining the North and South Pole is called the longitude line.
- The length of longitude lines is equal.
- The distance between longitudes increases as we move from the poles towards the equator.
- It takes 4 minutes for the Earth to cover 1° longitude.
- The prime longitude (meridian) is known as the Greenwich line. This is the zero degree longitude line. Exactly opposite to this is the 180° longitude line.

(0^0) zero degree latitude (equator) and (0^0) zero degree longitude intersect each other in the 'Gulf of Guinea' located on the western part of Africa. It is located in the Atlantic Ocean.

- **The prime meridian passes through the following countries**

1. United Kingdom	4. Algeria	7. Togo
2. France	5. Mali	8. Ghana
3. Spain	6. Burkina Faso	9. Atlantic Ocean

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- International standard time is determined by the prime meridian.
- The 180^0 longitude line is known as the International Date Line.
- There is a difference of 180° longitude between the Greenwich Line and the International Date Line, that is, there is a difference of 12 hours.
- The shortest distance between two places on the globe does not lie on the International Date Line.
- To reach from one place to another in less time, a ship must travel relative to longitude.

Equator

- It is also called the 0^0 latitude line or the equator
- It passes through the following countries - Ecuador, Colombia, Brazil, Gabon, Republic of Congo, Democratic Republic of the Congo, Uganda, Kenya, Somalia, Maldives, Indonesia, Kiribati

Tropic of Cancer

- 23½° North latitude is called the Tropic of Cancer.
- It passes through the following countries - Taiwan, China, Myanmar, Bangladesh, India, Oman, United Arab Emirates, Saudi Arabia, Egypt, Libya, Niger, Algeria, Mali, Mauritania, Western Sahara, Bahamas, Mexico

Tropic of Capricorn

- $23½^0$ South latitude is called the Tropic of Capricorn.
- It passes through the following countries - Chile, Argentina, Paraguay, Brazil, Namibia, Botswana, South Africa, Mozambique, Madagascar, Australia

Summer Solstice

- On 21st June, the Sun is perpendicular to the Tropic of Cancer, this is called Summer Solstice.
- This day is the longest day in the Northern Hemisphere.

Winter Solstice

- On 22 December the Sun is perpendicular to the Tropic of Capricorn.
- This day is the longest day in the Southern Hemisphere.

Equinox

- This is the position of the Earth when the sun rays fall perpendicularly on the equator.
- Day and night are equal everywhere.
- On 23 September and 21 March, day and night are of equal length across the entire earth.
- The midnight sun appears north of the Arctic Circle in the Northern Hemisphere and south of the Antarctic Circle in the Southern Hemisphere. Between these circles and the poles, there are 24-hour days in summer.

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